TOPIC 11

# Common Tests and Choosing One

Two-sample and paired t-tests, chi-square, ANOVA, non-parametric alternatives, effect sizes and a decision guide

**WHAT YOU WILL LEARN**

- Choose the right test from the type of data and the question
- Run and interpret two-sample, paired, chi-square and one-way ANOVA tests
- Know when to switch to non-parametric tests
- Check assumptions and report effect sizes
- Write up a result correctly

## 1. A map of the common tests

Every test in this topic follows the six steps of Topic 10: hypotheses, α, assumptions, test statistic, p-value, decision. Only the **statistic** and its **reference distribution** change. Choose by asking: *what type of data do I have, and how many groups am I comparing?*

| Question | Data | Test | Statistic / distribution |
|---|---|---|---|
| Is one mean equal to a value? | Numerical | One-sample t-test | t, df = n − 1 |
| Do two **independent** groups differ in mean? | Numerical | Two-sample (Welch) t-test | t, Welch df |
| Did the **same** subjects change? | Numerical, paired | Paired t-test | t on differences |
| Do 3 or more groups differ in mean? | Numerical, one factor | One-way ANOVA | F |
| Do observed counts fit a model? | Categorical counts | Chi-square goodness of fit | $\chi^2$, df = k − 1 |
| Are two categorical variables related? | Contingency table | Chi-square test of independence | $\chi^2$, df = (r − 1)(c − 1) |
| Compare means but data are non-normal or ordinal | Numerical / ordinal | Mann-Whitney, Wilcoxon, Kruskal-Wallis | Rank-based |
| Is a proportion equal to a value? | Binary | One-proportion z-test | z |

The one-proportion z-test is worked in Topic 10, section 8.

## 2. Two independent samples: the Welch t-test

Compare the means of two **separate** groups (for example, two teaching methods). $H_0$: $\mu_1 = \mu_2$ (equivalently $\mu_1 - \mu_2 = 0$).

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

**Welch's version** does not assume equal variances and is the safe default. (The older **pooled** t-test assumes both groups have the same variance and combines them into one $s_{pooled}$, with $df = n_1 + n_2 - 2$; it is only right if that assumption holds, while Welch loses almost nothing when it does, so Welch is preferred.) This is the same standard error you met in Topic 07, and the same t-statistic whose confidence interval appeared in Topic 09.

> **WORKED EXAMPLE** **Two teaching methods (from Topic 09)**
> Method A: $n = 40$, $\bar{x} = 82.4$, $s = 9.1$. Method B: $n = 35$, $\bar{x} = 78.1$, $s = 10.3$. Two-sided test at $\alpha = 0.05$. $t = \mathbf{1.904}$, $p = \mathbf{0.0611}$. Since $p > 0.05$ we **fail to reject $H_0$**. This matches the Topic 09 interval $[-0.21, 8.81]$, which just includes 0. The effect size is Cohen's $d = 0.44$ (small to medium); the study may simply be too small to detect it.

## 3. Paired samples: the paired t-test

When the two measurements are on the **same units** (before and after, left and right) they are not independent. Compute the **difference** for each pair and run a one-sample t-test on the differences with $H_0$: $\mu_d = 0$. Pairing removes person-to-person variation, so it is usually far more powerful.

> **WORKED EXAMPLE** **Blood pressure before and after treatment**
> Eight patients' systolic pressure: before = [140, 152, 138, 147, 160, 155, 149, 143], after = [132, 147, 136, 140, 151, 150, 145, 139]. Differences (before − after) = [8, 5, 2, 7, 9, 5, 4, 4]; mean difference $\bar{d} = 5.50$, $s_d = 2.33$. Paired $t = \bar{d}/(s_d/\sqrt{8}) = \mathbf{6.677}$, $df = 7$, $p = \mathbf{0.0003}$, so we **reject $H_0$**. A 95% CI for the mean drop is [3.55, 7.45].

For comparison, ignoring the pairing with a two-sample test gives $p = 0.149$: the effect disappears into between-patient variation. Always use the paired test for paired data.

## 4. Chi-square tests for counts

Chi-square tests compare **observed counts** (O) with **expected counts** (E) under $H_0$.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Large $\chi^2$ means the data are far from what $H_0$ predicts. Requirement: every expected count should be at least about 5.

### Goodness of fit: is the die fair?

| Face | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Observed | 25 | 17 | 15 | 23 | 24 | 16 |
| Expected ($H_0$: fair) | 20 | 20 | 20 | 20 | 20 | 20 |
| $(O - E)^2/E$ | 1.25 | 0.45 | 1.25 | 0.45 | 0.80 | 0.80 |

Total $\chi^2 = \mathbf{5.00}$ with $df = 6 - 1 = 5$, $p = \mathbf{0.416}$. We **fail to reject $H_0$**: 120 rolls with this much variation are quite consistent with a fair die.

### Independence: are two categorical variables related?

Test whether the outcome depends on the group. Expected count for each cell = (row total × column total)/grand total, and $df = (\text{rows} - 1)(\text{columns} - 1)$.

| | Success | Failure | Total |
|---|---|---|---|
| New treatment | 60 | 40 | 100 |
| Standard treatment | 45 | 55 | 100 |
| Total | 105 | 95 | 200 |

Expected counts if treatment and success were independent: 52.5, 47.5 / 52.5, 47.5. $\chi^2$ = **4.511**, df = 1, p = **0.0337**, so we **reject $H_0$**: the success rate differs between treatments (60% versus 45%). (Software applies a continuity correction to 2×2 tables by default; here it moves p from 0.034 to 0.047, still below 0.05 but closer, so say which version you used.)

The four $(O - E)^2/E$ terms, one per cell: $(60 - 52.5)^2/52.5 = 1.071$; $(40 - 47.5)^2/47.5 = 1.184$; $(45 - 52.5)^2/52.5 = 1.071$; $(55 - 47.5)^2/47.5 = 1.184$. Their sum is 4.511, the $\chi^2$ above. The **continuity correction** (Yates) subtracts 0.5 from each $|O - E|$ before squaring, because counts are whole numbers but the $\chi^2$ curve is smooth; it makes the test slightly more cautious.

> **COMMON MISTAKE** **Association is not causation**
> A significant chi-square says the variables are **related**. Only a well-designed randomized experiment can support a causal claim.

# 5. One-way ANOVA: comparing three or more means

Running many pairwise t-tests inflates the false-positive rate (Topic 10). **Analysis of variance (ANOVA)** tests all the group means at once. $H_0$: $\mu_1 = \mu_2 = \ldots = \mu_k$; $H_1$: at least one mean differs.

The idea: compare the variation **between** group means with the variation **within** groups.

$$F = \frac{\text{variance between groups (MSB)}}{\text{variance within groups (MSW)}} = \frac{SSB/(k-1)}{SSW/(N-k)}$$

## The three steps, on a tiny data set

Here **k** is the number of groups and **N** is the total number of observations (all groups together). Take k = 3 groups of 4 values, so N = 12.

| Group | Values | Group mean |
|---|---|---|
| 1 | 4, 5, 6, 5 | 5.00 |
| 2 | 7, 8, 6, 7 | 7.00 |
| 3 | 9, 8, 10, 9 | 9.00 |

1. **Means.** Group means are 5.00, 7.00, 9.00; the grand mean (all 12 values) is 7.00.
2. **Sums of squares.** Between: $SSB = \Sigma\, n_j (\bar{x}_j - \bar{x})^2 = 4(5.00 - 7.00)^2 + 4(7.00 - 7.00)^2 + 4(9.00 - 7.00)^2$ = **32.00**. Within: $SSW = \Sigma\, (x - \bar{x}_j)^2$, each value measured from its own group mean = **6.00**.
3. **Mean squares and F.** df between = k − 1 = 2 and df within = N − k = 9. MSB = 32.00/2 = 16.00, MSW = 6.00/9 = 0.667, so F = 16.00/0.667 = **24.00**, p = 0.00025.

Now a larger example.

| Group | Scores | Mean |
|---|---|---|
| A | 78, 82, 85, 80, 79, 84 | 81.33 |
| B | 72, 75, 70, 74, 77, 73 | 73.50 |
| C | 80, 83, 79, 85, 81, 78 | 81.00 |

| Source | SS | df | MS | F | p |
|---|---|---|---|---|---|
| Between groups | 235.44 | 2 | 117.72 | 17.17 | 0.00013 |
| Within groups | 102.83 | 15 | 6.86 | | |
| Total | 338.28 | 17 | | | |

$F(2, 15) =$ **17.17**, $p =$ **0.00013**, so we **reject $H_0$**: the means are not all equal. ANOVA does not say *which* differ; a **post-hoc test** does (Tukey HSD controls the overall error rate): A vs B p = 0.0003, A vs C p = 0.9736, B vs C p = 0.0005. Group B is lower than both others, while A and C are not clearly different.

Assumptions: independent observations, roughly normal groups, similar variances. ANOVA is fairly robust with equal group sizes.

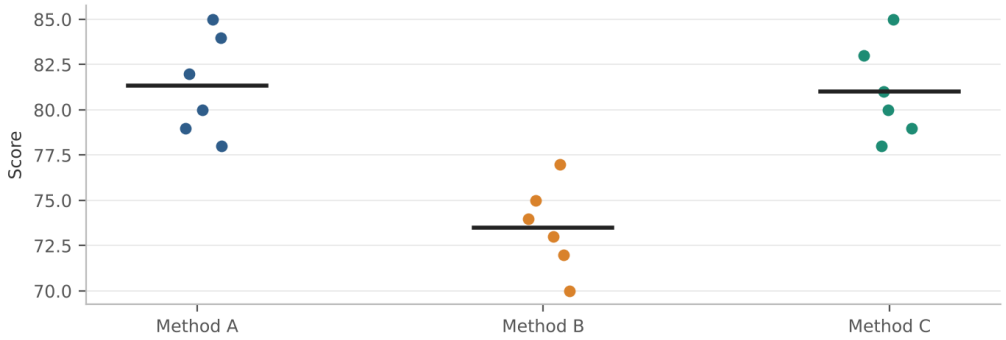



Figure 1. Scores by group; black bars are group means. The spread within each group is small compared with the gap to group B, giving a large F.

## 6. When the assumptions fail: non-parametric tests

The t-tests and ANOVA assume roughly normal data (or large n). With small samples of skewed or ordinal data use **rank-based** alternatives. They compare ranks rather than raw values, so outliers matter much less, at a small cost in power when the data really are normal.

| Parametric test | Non-parametric alternative | Compares |
|---|---|---|
| One-sample / paired t-test | Wilcoxon signed-rank test | Median of differences |
| Two-sample t-test | Mann-Whitney U (Wilcoxon rank-sum) | Whether one group tends to be larger |
| One-way ANOVA | Kruskal-Wallis test | Groups' rank distributions |
| Pearson correlation | Spearman rank correlation | Monotonic association |

### The rank idea on 4 + 4 values

Pool all 8 values, rank them from smallest (1) to largest (8), then add up the ranks within each group. If the groups were alike, the rank sums would be about equal.

| Group | Values | Ranks | Rank sum |
|---|---|---|---|
| A | 12, 15, 11, 18 | 2, 4, 1, 6 | 13 |
| B | 20, 17, 25, 14 | 7, 5, 8, 3 | 23 |

The rank sums are 13 and 23 (they always total 36). Group B's values rank higher. Mann-Whitney U is computed from these sums; here U = 3 and p = 0.200. Only the order of values matters, not their size.

- **Wilcoxon signed-rank:** rank the sizes of the paired differences, then add up the ranks of the positive ones and the negative ones.
- **Kruskal-Wallis:** rank all observations together, then compare the average rank of each of the 3+ groups (ANOVA on ranks).
- **Spearman:** rank each variable separately, then take the ordinary correlation of the two sets of ranks.

Example: two small skewed samples (the first has an outlier at 12.0). Welch's t-test gives p = 0.228, while Mann-Whitney U gives p = 0.026. The outlier inflates the variance in the first group and weakens the t-test; ranks are not fooled by it.

## 7. Assumptions and effect sizes

| Test | Key assumptions | How to check |
|---|---|---|
| t-tests | Independence; roughly normal data or n large; (Welch: no equal-variance need) | Histogram, Q-Q plot, outlier check |
| ANOVA | Independence; normality in each group; similar variances | Residual plots, Levene's test |
| Chi-square | Independent observations; expected counts ≥ 5 | Look at the expected table |

A p-value says whether an effect is detectable; an **effect size** says how big it is.

| Effect size | For | Rough guide |
|---|---|---|
| Cohen's d = $(\bar{x}_1 - \bar{x}_2)/s_{pooled}$ | Two means | 0.2 small, 0.5 medium, 0.8 large |
| $\eta^2$ = SSB/SST | ANOVA | 0.01 small, 0.06 medium, 0.14 large |
| Cramér's V | Chi-square tables | Depends on table size |
| Risk difference / odds ratio | Binary outcomes | Context-specific |

For the ANOVA above, $\eta^2 = 0.70$: about 70% of the variation in scores is explained by method, a large effect.

**DEFINITION** **Terms you will meet**

- **Tukey HSD:** a post-hoc test after ANOVA that compares every pair of means while keeping the overall error rate at α.
- **Levene's test:** tests $H_0$ that groups have equal variances (a check before ANOVA or a pooled t-test).
- **Cramér's V:** effect size for a chi-square table, from 0 (no association) to 1 (perfect).
- **Odds ratio:** odds of success in one group divided by odds in another, where odds = p/(1 − p); 1 means no difference.
- **Fisher's exact test:** tests a 2×2 table exactly, used when expected counts are small (below 5).
- **Pooled standard deviation** (the s_pooled in Cohen's d), the average spread of two groups weighted by their sizes:

$$s_{pooled} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

## 8. Writing up a result

**KEY IDEA** **A good report contains**

- The test used and why.
- The statistic with its degrees of freedom and the p-value.
- An effect size and a confidence interval.
- A conclusion in plain language about the original question.

**WORKED EXAMPLE** **Example**

'A paired t-test showed that systolic blood pressure fell significantly after treatment (mean drop 5.5 mmHg, 95% CI [3.6, 7.4], t(7) = 6.68, p < 0.001, n = 8).'

## 9. A decision guide

| Start here | Then | Use |
|---|---|---|
| Outcome is numerical | 1 group vs a value | One-sample t-test |
| | 2 independent groups | Welch t-test (Mann-Whitney if non-normal) |
| | 2 paired measurements | Paired t-test (Wilcoxon if non-normal) |
| | 3+ groups | One-way ANOVA (Kruskal-Wallis if non-normal), then post-hoc |
| Outcome is categorical | 1 variable vs a model | Chi-square goodness of fit (or z-test for one proportion) |
| | 2 variables, table of counts | Chi-square test of independence (Fisher's exact if counts are small) |

## 10. In Python

```
from scipy import stats
import numpy as np

# Two independent groups
stats.ttest_ind(a, b, equal_var=False)            # Welch t-test
stats.mannwhitneyu(a, b)                          # non-parametric version

# Paired
stats.ttest_rel(before, after)
stats.wilcoxon(before, after)

# Categorical
stats.chisquare(observed, expected)               # goodness of fit
stats.chi2_contingency(table)                     # independence test

# 3+ groups
stats.f_oneway(g1, g2, g3)
stats.kruskal(g1, g2, g3)
stats.tukey_hsd(g1, g2, g3)                       # post-hoc comparisons
```

## 11. Where to go next

This handbook ends where modelling begins. Natural next topics are **correlation and linear regression** (testing relationships between numerical variables, using the same t and F ideas), **logistic regression** for binary outcomes, **power analysis and experimental design** (A/B testing), **non-parametric and resampling methods** (bootstrap, permutation tests) and **Bayesian inference**, which treats parameters as random and answers 'what is the probability that $H_0$ is true?' directly.

**SUMMARY**

- Choose the test from the data type and the design (one group, two independent, paired, three or more, counts).
- Welch t-test for two independent means; paired t-test for before/after; ANOVA for three or more means.
- Chi-square compares observed with expected counts: goodness of fit and independence.
- Rank-based tests are the safe choice for small, skewed or ordinal data.
- Always report effect size and confidence interval, not just the p-value.

## Practice questions

Try each question before reading the answer underneath it.

**Q1.** Fifty customers rate satisfaction before and after a redesign. Which test?

> **Answer.** Paired t-test (or Wilcoxon signed-rank if the differences are far from normal), because the same customers are measured twice.

**Q2.** Mean delivery time is compared across 4 couriers. Which test and why not 6 t-tests?

> **Answer.** One-way ANOVA. Six pairwise tests inflate the overall false-positive rate (about $1 - 0.95^6 = 26\%$).

**Q3.** A 3 × 2 table has expected counts all above 5. Give df for a chi-square test of independence.

**Answer.** $(3 - 1)(2 - 1) = 2$.

**Q4.** A two-sample t-test with n = 3,200 per group gives p = 0.045, but Cohen's d = 0.05. Comment.

**Answer.** Statistically significant ($p < 0.05$) but the effect is tiny ($d = 0.05$, far below even a 'small' 0.2), so it is probably not practically important. The large n made it detectable.

**Q5.** ANOVA gives p = 0.001. Can you say all means differ?

**Answer.** No, only that at least one differs. Follow with a post-hoc test such as Tukey HSD.

**Q6.** Two small samples are heavily skewed with outliers. Which test would you use for a difference?

**Answer.** Mann-Whitney U, since it is rank-based and robust to outliers.